



**National Computing Education Accreditation
Council**

NCEAC



NCEAC.FORM.001-D

COURSE DESCRIPTION FORM: CS-4001 PROFESSIONAL PRACTICES in IT
COURSE DESCRIPTION FORM

PROGRAM TO BE EVALUATED BS-CS-FALL 2025

Course Code	CS-4001		
Course Title	Professional Practices in IT		
Credit Hours	3+0		
Prerequisites by Course(s) and Topics	Nil		
Grading Policy	Absolute grading		
Policy about missed assessment items in the course	Retake of missed assessment items (other than midterm/ final exam) will not be held. For a missed midterm/ final exam, an exam re-take/ pre-take application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee will decide the exam re-take/ pre-take cases. - Late policy: 50% deduction after 48 hours. No awards whatsoever after ONE WEEK after the deadline. - Plagiarism punishment up to 20 weightages.		
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	Assignments/ Quizzes	8	(4 Assessments)
	Class Participation	4	(4 Assessments)
	Midterm Exam	2	(15 each)
	Project	1	(8 Marks)
	Final Exam	1	(50 Marks)
Course Coordinator	Anaum Hamid.		
URL (if any)			
Current Catalog Description	This course allows students to infer and analyze the Professional situations with the power of character and professionalism, Computing Ethics, Philosophy of Ethics. The Structure of Organizations, Finance and Accounting, Anatomy of a Software House, Computer Contracts, Intellectual Property Rights, The Framework of Employee Relations Law and Changing Management Practices, Human Resource Management and IT, Health and Safety at Work, Computer Misuse, Regulation and Control of Personal Information. Overview of the BCS Code of Conduct, IEEE Code of Ethics, ACM Code.		
Textbook	1. Professional Issues in IT by Frank Bott 4th Edition 2. Ethical and Social Issues in Information Age, Kizza J. M., 5th Edition (2013), Springer-Verlag		



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Course Learning Outcome	Course Learning Outcomes (CLO): Upon successful completion of the course, the students will be able to		
	CLO1	Ability to engage in informed critical reflection on the nature of professionalism and ethical challenges inherent in professionalism	
	CLO2	Ability to bring to bear ethical analysis and reasoning in the light of normative ethics frameworks on a selection of ethical challenges and dilemmas across the chosen range of professions	
	CLO3	Analyze the project management techniques for IT projects to initiate, plan, execute and evaluate a project and Startups.	
	CLO4	Present strategies for gaining confidence in managing projects through simple project planning and working in a team to create a project plan which includes key tasks, critical path, dependencies and a realistic timeline.	
Program Learning Outcome	Program Learning Outcomes (PLO): Upon successful completion of the course, t students will be able to		
	PLO-1	Computing Knowledge	Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems.
	PLO-2	Problem Analysis	An ability to identify, formulate, research literature, and analyze Complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
	PLO-3	Design/Develop Solutions	An ability to design solutions for complex computing problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
	PLO-4	Investigation & Experimentation	Conduct investigation of complex computing problems using research-based knowledge and research-based methods.
	PLO-5	Modern Tool Usage	Create, select, and apply appropriate techniques, resources and modern computing tools, including prediction and modelling for complex computing problems.
	PLO-6	Society Responsibility	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to context of complex computing problems.



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WEEK	TOPICS/SUBTOPICS	Credit Hours	CLO'S
01	Introduction to Professional Practices • Introduction of Professional practices. • Rights. Negative vs positive rights. • Legal Rights vs Moral rights • Character, morals, personality, culture, self-identity, Professional Ethics • Introduction to Information ethics, types of ethics and stages of ethics in information systems, Standards in ethical thinking, • profession and role of collective body, course accreditation • Historic review of information ethics, key contributors, knowledge areas for information related ethics. • Professionalism in Business, Self-Professionalism • Ethical Decision-Making scenarios: Case studies for Public, Client and employer, Product, Judgment, Management, Profession, Colleagues, Self, professional responsibilities in context of code of conduct • Work from home	3 Hours	1,2
02	Organizations: (Kinds of organization, registrations, legal formal framework) • Open, closed, hierarchy vs flat structure, Google/Facebook • Organizations, Commercial organization, corporates • Private limited companies Constitution of limited companies (the <i>memorandum of association</i> and <i>articles of association</i>) • Directors • Noncommercial bodies • How to start a company.	2 Lectures + 1 Lecture (Guest Speaker)	2,3
04	Good and Bad (Google vs Seith Organization) • Your Startup • Seith Organization • Google/Facebook Culture of Organization	2 Lectures Theory 1 Guest Speaker	2,3
	• Decision making, independence, micro/macro management, ownership, talent acquisition • Remote Teams (Work from home) • Corporate social responsibility		
05	MID 1		



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06	Professional Contracts - Focus areas for contract terms and conditions, what is to be produced, what is to be delivered, ownership of rights, confidentiality, mode of payments, A model to calculate payments for delays and changes, penalty terms and conditions, obligations of the client, standards and methods for working, - Progress meetings, parties nominated focal persons, Acceptance procedure, Warranty and maintenance clauses Effect of Inflation for long, force de majeure circumstances, termination of the contract, Arbitration nomination, Applicable laws, Contract hire Time and material, consultancy contract, liability for the defective software programs. - NDA, MOU, Proposals, Legal Issues (demonstrating NDA & MOU) - Software related contracts, Legal value of contract for software development, professional practices for writing quality software contracts, structure of fixed price contracts for custom built software programs - Risk	3 Theory Lectures + 1 hr Guest Speaker	3,4
07	Human Resource Management and health/ safety of employees - Introduction - Model of human resource management, features of HRM - Training of employees, keeping track of skill of employees, problems in organization faced by HRM. - Health and safety of employee is a major concern - Act of 1974 related to health and safety - Employee/Employer contracts, HR activities, appraisals, discrimination, Harassment	2 hrs Lecture Theory + HR Guest Speaker + Scaling a solo startup to SMEs Guest Speakers	1,2,3,4
08	Intellectual Property and copyright laws - IP Principles and Laws basic definitions and concepts, Key challenges for protection of IP rights in new technologies. - Ethical Arguments About Copying software products, common causes of IP violations in society - Forms of IP, Patents, trademarks and copyright laws, applications in society and social aspects - Case studies for applications of IP violations - Plagiarism - Demonstration of Patent Document, How to write a patent	2 hrs Lectures Theory + 1 Guest Speaker	2,3
09	MID 2		
10	Discrimination Ethnic, racial, gender, age. Perspectives (perceptions), Judgement Anti discrimination legislation Direct/Indirect Positive/Negative (Lawful/unlawful) Professional discrimination Nepotism		1,2,3



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	Disability (For blinds, autistics, deafs. etc)		
11	Service Oriented Organizations • Legal liability and mechanism, safety related applications • Regularity issues, standards, certificate and licensing, professional code of practices, regulation by law. • Product liability and consumer protection act (CPA) 1987, negligence • Work from home, human/professional perspective, • Risk	2 hrs Theory + 1 hr Guest Speaker 2 page note CP 2 Marks Quiz 'Performance during work from home' System HR (tentative)	2,3,4
12-13	Cyber Law, computer misuse • Categories of misuse • computer frauds, unauthorized access to a computer • Computer misuse, hacking, types of hacking • Tools of hacking, computer security • Ethical hacking, penetration testing • Hacker code of conduct Data Protection Data protection and privacy of data importance. • Impact of internet, factors affecting regulation • Defamation and protection • Key weakness in code of conduct, future challenges • Defining privacy, Common privacy threats, New Technologies rising risks, managing personal data, Critical need for privacy protection. • Digital Frauds • Risks	3 Theory Lectures 2 hrs Guest Speaker 1. Careem 2. Financial Sector (IS Audit Professional) 1 hr article review and CP 2 Marks Quiz	2,3
14	Internet Issues • Information security basic concepts, CIA principles • Information classification, IS organizational structure • Security plans and case studies • Cyber Bullying • Defamation • Fake content/Rumours/Digital Blackmailing • Cyber Crime Legistelation in Pakistan • Social platform Tik Tok, Youtube • Risks	3 hrs Theory Lectures	2,3